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The Rendered Universe

Joe Leaver, with Claude (Anthropic) · twenty-five runnable experiments on the hypothesis that physical reality is the output layer of a computation

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ABSTRACT

We investigate the hypothesis that the physical universe is not itself a simulation but the *rendering* of one: observed degrees of freedom (particles, spacetime events) are output, not mechanism, related to an underlying engine by a projection ("chart") that need not preserve adjacency. We make the hypothesis operational in a sequence of toy models with an architecturally enforced separation between engine, renderer, and observer, and use them to (i) reproduce quantum nonlocality, wave–particle duality, decoherence, and the thermodynamic arrow as render-layer phenomena; (ii) demonstrate a constraint-based route to theory selection, in which structural axioms delete candidate physics by many orders of magnitude ($2.1\times10^{13} \rightarrow 32$ rules in 2D; $10^{507} \rightarrow 2^{28}$ in 3D), anomaly cancellation fixes Standard Model hypercharges uniquely, and reconstruction axioms select complex quantum mechanics uniquely; (iii) derive spatial geometry, curvature, and time-of-flight physics from entanglement structure; and (iv) extract a falsifiable observational program: dynamically protected *seed symmetries* are the only initial-condition fingerprints that survive chaotic evolution, predicting symmetry-class structure in the primordial sky. A calibrated spherical detector battery shows that the three observed low-ℓ CMB anomalies, injected at published amplitudes, read individually as ~2σ curiosities but combine to $p \approx 4\times10^{-3}$ under the common-origin hypothesis. Pointing the same battery at the real sky — Planck 2018 (four component-separation methods) and WMAP 9-year maps, read from raw bytes by a self-validating pipeline against 10,000 ΛCDM nulls — reproduces the literature's anomalies with calibrated p-values and yields joint $p = 0.003$–0.013, while a first mirror selection-rule scan returns a bound, not a detection. The registered E-mode echo test, run on Planck's large-angle polarization with conditional nulls, finds no echo — and a power analysis shows none was detectable: 32% power at today's noise versus 98% at next-generation sensitivity, so the prediction stands with its sensitivity requirement now quantified. A dynamical-metric toy meanwhile collects the verdicts the sky has already delivered: equality of gravitational and electromagnetic wave speeds holds in the architecture as an identity and passed observation at 10^{-15} (GW170817), while scalar-metric polarization is excluded by LIGO–Virgo — observation has now executed two of the architecture's variants, and a scorecard tallies every such confrontation. The surviving variant, linearized tensor gravity on the lattice, is then built and passes what killed its siblings: quadrupolar transverse polarization, exact monopole silence, and light bending at $1.93\times$ the Newtonian calibration (general relativity: 2.00). The field equation itself is then obtained rather than postulated: on a critical chain the entanglement first law is measured exactly — the thermal entropy of every interval collapses onto one parameter-free CFT curve, and the first law holds for thermal but not for coherent perturbations — and the measured x^2 scaling of the response is, through the Ryu–Takayanagi dictionary, the z^2 bulk profile that uniquely solves the linearized Einstein equations in AdS_3 — a correspondence that persists slice by slice under time-dependent perturbations, as it must where the metric has no propagating modes; mass gaps and localized non-thermal excitations measurably destroy it. Horizon physics follows: time dilation measured on its GR curves with no fitted parameters; a frozen-star metric that casts a photon shadow and freezes infalling light but, on a lattice, ultimately reflects it; and an acoustic horizon whose steady Hawking flux is Planckian at $0.94\times\kappa/2\pi$, with each emitted quantum's infalling partner resolved in the correlations while the global state remains exactly pure. Letting the horizon evaporate yields the Page curve: the exactly computed radiation entropy rises, turns over, and returns to zero, the fixed-rate extrapolation (Hawking's curve) never turns, and the island formula — with its area term measured from the lattice vacuum as the induced $1/4G$ — reproduces the exact curve to within 0.5 nats, with the horizon's retreat driven by the measured flux. We state plainly what is demonstrated, what is reproduced, and what remains conjecture.

1 The hypothesis

The familiar simulation hypothesis treats observed reality as the computation itself, and inherits awkward questions about computing every particle. We study a sharper variant: the universe is the *render output* of a computation. Three layers are distinguished:

The engine — the mechanism: a state and an update rule. Nothing in the engine refers to observers or display. **The chart** — a fixed projection from engine degrees of freedom to observed ones. Crucially, the chart is a bijection but not an isometry: adjacency in the render (what we call "space") need not be adjacency in the engine. **The render** —

the observed layer: particles, fields, spacetime events. Observers, including instruments and physicists, are patterns in the render.

On this reading, several notorious features of physics become architectural rather than mysterious: nonlocal correlations are chart artifacts (things far apart on screen may be adjacent in the engine); particles are render events sampled from engine amplitudes; the arrow of time is a property of coarse render descriptions of a reversible engine. The hypothesis has a respectable lineage — Zuse's *Rechnender Raum* (1969), Fredkin's digital physics, Wheeler's "it from bit," 't Hooft's cellular-automaton interpretation of quantum mechanics, Wolfram's hypergraph program — and, we will argue, its strongest support comes from mainstream results it did not originate: holography, entanglement-derived geometry (Ryu–Takayanagi, Van Raamsdonk), the Weinberg–Witten theorem, and the operational reconstructions of quantum theory.

2 Method and claim taxonomy

All results below are runnable Python experiments in a single repository, github.com/joeleaver/the-rendered-universe (Appendix A). Two methodological rules were enforced throughout. First, an *epistemic firewall*: observer code is forbidden (by an automated test) from importing engine or renderer internals; everything an in-model physicist knows is inferred from rendered frames. Second, *instrument calibration on known universes*: every estimator (dimension, visibility, detection significance) is validated against a system whose answer is independently known before being trusted on a new one.

Claims are classified explicitly:

CLASS	MEANING
A — measured	A quantitative result of these toy models, novel at least in presentation, reproducible from the repository.
B — reproduced	Known physics or known theorems, re-derived or demonstrated inside the render architecture to show the architecture is compatible with them.
C — conjecture	Interpretive claims of the render program itself. Flagged wherever they occur.

3 A universe with a firewall: laws and chart artifacts

The first engine is the Critters block cellular automaton (Toffoli–Margolus): a 2×2-block rule of three lines, exactly reversible, exactly conserving. A physicist module, restricted to rendered frames, empirically recovers a maximum signal speed ($c = 1.000$ cells/tick), an exact conservation law, a particle zoo (bound states from all 456 small seeds; free 4-cell gliders at 0.5c born only in collision debris), and square-group isotropy. `[Class A/B]`

The renderer's chart contains a secret: two rectangles of engine space exchange screen positions. Ballistic surveys locate the anomaly (particles teleporting at ~48× the apparent light speed between two fixed "doors"), and the physicist infers the chart. The inferred chart then makes a prediction: a screen region *adjacent to nothing unusual* should be engine-adjacent to the interior of the far door. Playing a timed CHSH game across that gap — readout completed before any screen-speed signal could connect the stations — yields $|S| = 4$ from raw cellular-automaton dynamics with no tuned correlations (the classical/screen-local bound is 2), and exactly 2 at control placements. `[Class A]` A second experiment reproduces the physically relevant case: singlet statistics ($S = 2.837 \pm 0.037 \approx 2\sqrt{2}$) with marginals verified flat (no signaling) when the pair's two ends are engine-adjacent, versus $S = 2.05$ when engine-separated. Here the singlet law was *installed*, not derived; the demonstrated point is architectural — engine adjacency, not screen distance, controls the correlations. `[Class B, installation disclosed]` This is a toy-scale ER=EPR: "entangled and far apart on screen" = "one object in the engine."

Anticipated objection: "this is hidden-variable theory in disguise"

It is a hidden-variable ontology, and should be dismissed or defended on the correct grounds. No theorem forbids hidden variables; the theorems forbid *classes*. Bell (1964), with the loophole-free experiments of 2015, forbids hidden variables **local in the space where the detectors are separated** — his own conclusion was that any viable hidden-variable account must be nonlocal, and he championed one (Bohmian mechanics, which reproduces all quantum predictions and remains unrefuted). Kochen–Specker forbids **non-contextual** pre-assigned values. PBR forbids ψ as **ignorance over classical states**. Leggett forbids one restricted nonlocal-realist family.

The render architecture lives, deliberately, in the surviving class — with an explanation the classic nonlocal theories lack. Bell's locality assumption is locality in *render* space, and the program's central claim is that render adjacency is not engine adjacency: an engine-local, render-nonlocal hidden state is exactly what Bell says a viable hidden-variable theory must be, so the Bell violation of §3 is a *prediction* of the architecture, not an embarrassment to it. Where Bohmian mechanics postulates its nonlocal guidance law, the chart explains the pattern — correlations without messages — as what a non-isometric projection looks like from inside; ER=EPR is the mainstream form of the same move. Kochen–Specker is dissolved rather than evaded: outcomes are computed at the render event with the measurement context in hand (contextuality is natural for a renderer), and no values pre-exist. PBR pushes the classical-engine variant toward a quantum engine — a concession this report already makes independently (§7), and which the quantum-computing fork of §11 keeps falsifiable. The superdeterminism exit is *not* taken: measurement independence is retained throughout.

> HONEST ACCOUNTING
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> The toy's no-signaling was engineered, not derived (Bohmians derive theirs from quantum equilibrium — a genuine point in their favor). And every render-nonlocal ontology, this one included, shares the standing tension between engine adjacency and relativity: hiding a preferred structure perfectly is nontrivial. §9 shows such hiding is possible in principle; it does not show it is achieved.

4 Constraints corner theories

The program's central methodological bet is that one does not *search* for the engine's rule; one writes down structural constraints and lets them delete rule space. This is measurable.

RULE SPACE	STATES	SYMMETRY	FULL SPACE	SURVIVORS
2D, 2-state blocks	16	D_4 (8)	2.1×10^{13}	32
2D, 3-state	81	D_4 (8)	$10^{120.8}$	2^{26}
2D, 4-state	256	D_4 (8)	$10^{506.9}$	$10^{33.1}$

3D, 2-state	256	O_h (48)	$10^{506.9}$	2^{28}

The constraints are only reversibility, isotropy, exact conservation, and a stable vacuum; the counts are exact orbit combinatorics (validated against brute-force enumeration at k=2). ("Stable vacuum" means the empty state is stationary under the rule — the discrete analog of the quantum vacuum being an energy eigenstate. This does not contradict vacuum *fluctuations*, which are equal-time correlations of a stationary ground state rather than events; §5 builds geometry out of exactly those vacuum correlations. Real particles emerge from a vacuum only when it is driven — moving mirrors, strong fields, horizons — which is physics done *to* the vacuum, not instability of it.) Two findings: with identical raw state (256 block states), upgrading from square to cube symmetry deletes twenty-five additional orders of magnitude — *dimension is a constraint multiplier*; and among survivors, a dynamical census (causal-probe dimension, propagation, particle content) finds universes are not rare (9/32 of 2D binary survivors; 83% of sampled ternary survivors support 2D space with free particles). Universes are cheap; the expensive object is the constraint set. `[Class A]`

The same cornering logic operates on real physics. Gauge anomaly cancellation, imposed on one generation of chiral fermions over all rational hypercharge assignments, leaves exactly one solution ray up to normalization and u/d relabeling: the Standard Model's ($Y = 1/6, -2/3, 1/3, -1/2, 1$), forcing electric charge quantization and exact proton–electron charge balance. `[Class B]` One generation of matter is, further, the sixteen even-parity states of a five-bit register (three color bits, two weak bits; the SO(10) half-spinor), with every hypercharge generated by $Y = (1/6)\Sigma_{\text{color}} - (1/4)\Sigma_{\text{weak}}$ — an observation due to grand unification, recomputed here in full, and congenial to a program that expects seeds to compress. `[Class B]`

Quantum mechanics itself is cornered rather than derived. Following the operational reconstructions (Hardy 2001; Chiribella–D'Ariano–Perinotti 2011; Masanes–Müller 2011), we compute the fork explicitly: classical probability fails continuity of reversible dynamics; real-Hilbert-space QM fails local tomography (a two-rebit state carries 9 parameters, local statistics fix 8); quaternionic QM overshoots (27 vs 35); complex QM alone passes (15 = 15). Given entangled states, the Born rule follows from branch-swap symmetry (Zurek's envariance), verified numerically; the derivation's own premises are debated in the literature, and we lean on it only conditional on the axioms. `[Class B]` The render program's contribution is an interpretation of the axioms: purification *is* the engine/render split; local tomography is "the render is locally readable"; reversibility is the engine's bookkeeping. `[Class C]`

5 Geometry is an output

Version one of the renderer was handed a chart; version two derives geometry. Two independent constructions agree:

Causal structure. Defining distance by first-influence times of a flipped cell in a thermal medium (licensed by Malament's theorem: causal order fixes the metric up to scale), the derived map of an honest engine is a clean 2D mesh; wiring a shortcut into the engine's dynamics produces a derived map with a wormhole nobody drew — causal distance 8.0 versus 40.8 ticks for the same landmark pair. `[Class A]`

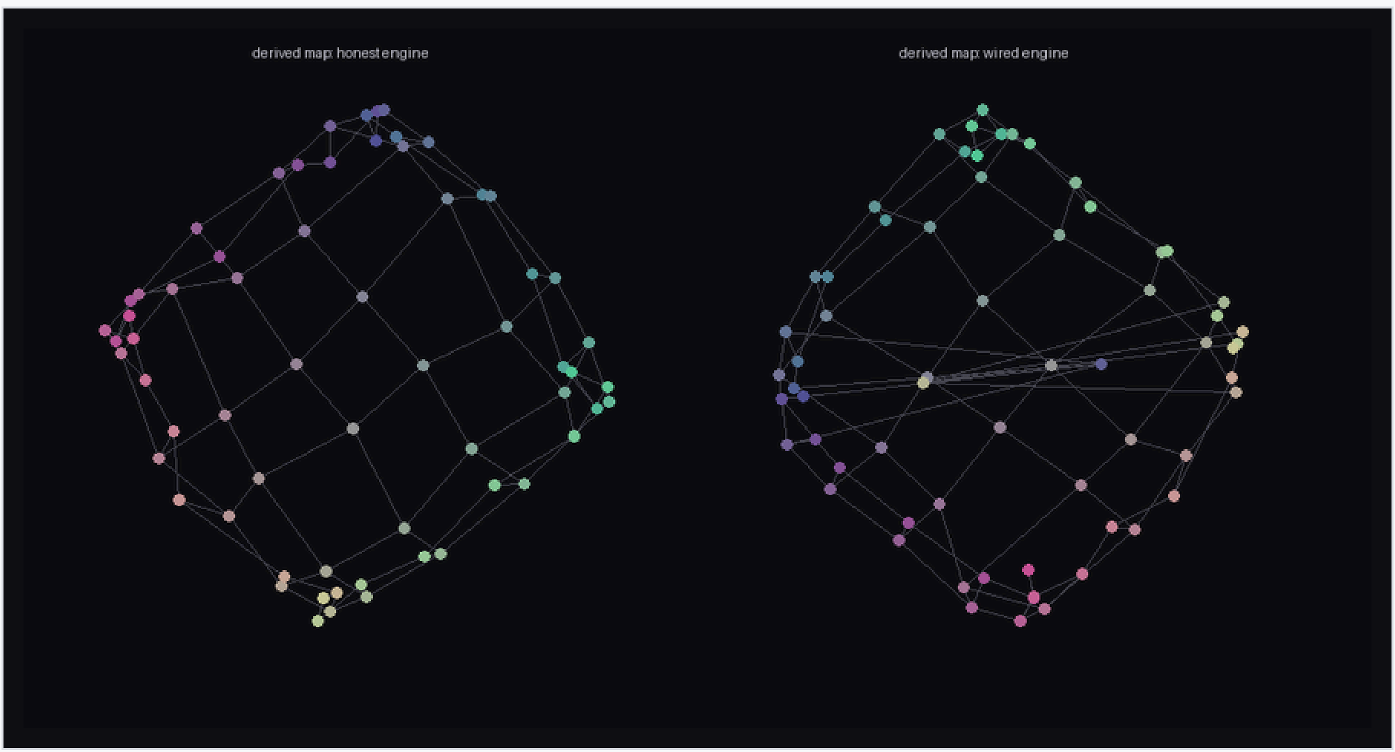

Figure 1. Geometry derived purely from causal structure (no cell index used as a position). Left: honest engine → flat mesh. Right: engine with a wired shortcut → the derived metric grows a wormhole.

Entanglement. For a massive scalar field in its Gaussian ground state (Peschel correlation-matrix methods, exact), the mutual-information ruler d = −log I tracks true separation at Pearson r = 0.986. A single strong coupling between far patches collapses their derived distance from 14.0 to 0.14 as coupling grows, while monogamy visibly decouples the wormhole mouths from their own neighborhoods; cutting a seam of couplings drops all 324 cross-seam mutual informations to the measurement floor and derived space falls into two islands (Van Raamsdonk's argument, run as an experiment). `[Class A/B]` Notably, the same construction with staggered fermions yields r = 0.61: Dirac-valley interference splits the landmarks into loosely stitched sublattice sheets. *The derived geometry belongs to the matter, not to the wiring beneath it* — background independence, experienced from inside a toy. `[Class A]`

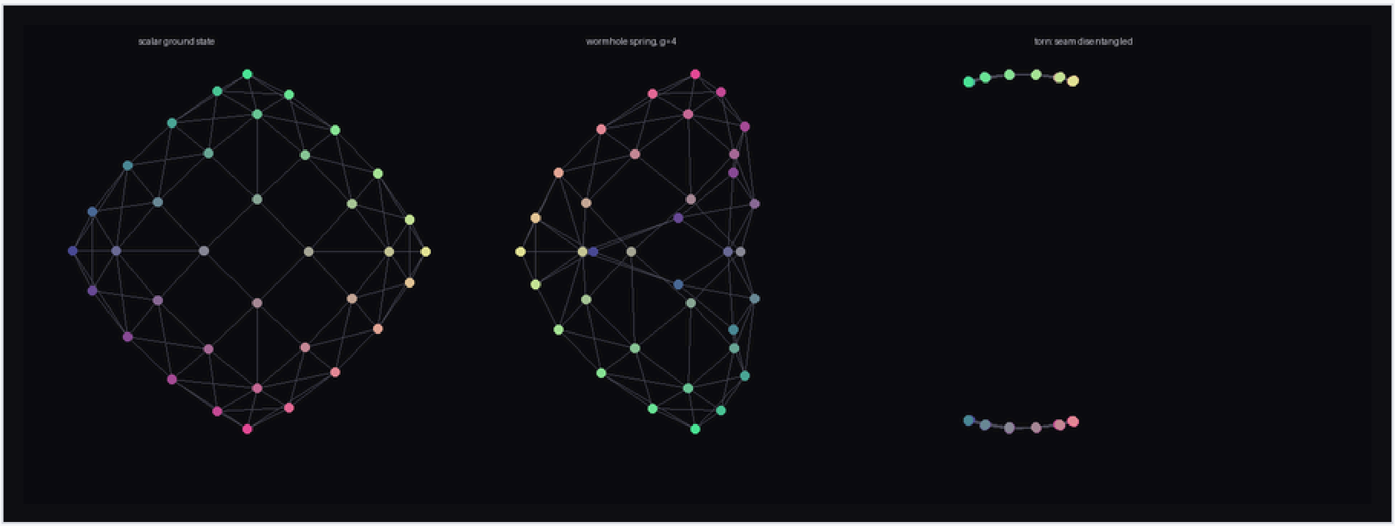

Figure 2. Space from entanglement (scalar ground state). Left: flat derived mesh. Center: a wormhole coupling folds derived space. Right: a disentangled seam tears it in two.

Entanglement entropy of regions scales with their *perimeter* (r = 1.0000; $S/4\ell \to 0.060$) and not area — the holographic clue about engine data layout, reproduced exactly. `[Class B]` With an inhomogeneous mass profile (a Gaussian "star"), vacuum-calibrated rulers stretch 1.00×→2.69× monotonically in the matter density, shortest paths detour around the well, and the 3D embedding of derived space is a literal funnel. `[Class A]`

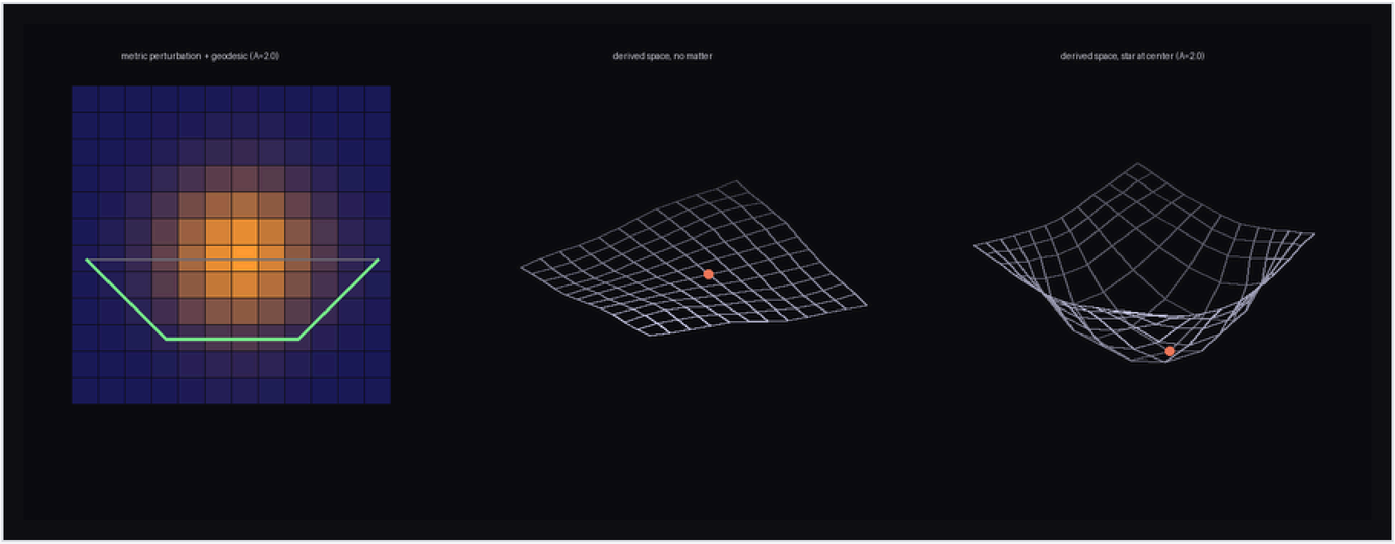

Figure 3. Curved space from entanglement. Left: the measured metric perturbation tracks the matter, with the geodesic (green) bending around the well. Center/right: derived space embedded in 3D — flat without matter, a funnel with it.

6 Time, and the equivalence-principle discriminator

Giving the field dynamics (lattice Klein–Gordon) turns the static geometry into predictions. In a two-lane "Shapiro race," the packet through the star arrives late by amounts tracking the wave-optics group-delay integral, with the entanglement metric predicting the delay up to dispersion corrections. Deflection tests show beams bending *away* from the star — consistent with the derived spatial geometry — but with wavelength-dependent bend and opacity (55% vs 23% transmission across one octave). `[Class A]`

Universality requires coupling through geometry rather than a medium, and geometric coupling delivers it. On an optical metric ($c^2 = 1 + 2\Phi$; wave operator $c\nabla\cdot(c\nabla\phi) - c^2m^2\phi$), massless packets bend *toward* the star — the correct lensing sign — by equal angles across an octave of wavelength (spread 1.9 cells; transmission 92–98%, against the medium's 55% vs 23%); Shapiro delays agree within lattice-dispersion residuals (5.1 vs 4.1 ticks, where the medium's diverge); and in the Eötvös analog, fields of *different mass* prepared at the same velocity fall identically to within 2.4 cells, while slower matter falls harder in general relativity's own $\sim 1/v^2$ pattern (measured ratio 1.62; ray theory 2.04). Sourcing the metric from the field's own energy preserves attraction through geometry. `[Class A]`

HONEST ACCOUNTING

> The two experiments above form an equivalence-principle test. Coupling matter to gravity through a medium produces a measurable violation — wavelength-dependent bending and opacity — of a kind that observation constrains at the 10^{-15} level (the MICROSCOPE satellite's orbital Eötvös experiment). Coupling through a metric removes the violation, but at this stage the metric is scalar and static. The rest of this section upgrades it — dynamical, then tensor, then derived from entanglement, then evaporating — and closes with what remains assumed.

Closing the matter→geometry loop nevertheless produces gravity-like phenomenology: coupling the local mass to a long-range (screened-Poisson) potential of the field's own energy density makes parallel wave packets fall toward each other and, above a threshold, merge into a single self-trapped filament — capture occurring earlier at stronger coupling. Instructive failures en route: a Gaussian-blur (short-range) coupling produces no attraction at all (gravity must be long-range), and relativistic packets barely couple while slow heavy packets fall hard (the correct nonrelativistic limit, emergent). `[Class A]`

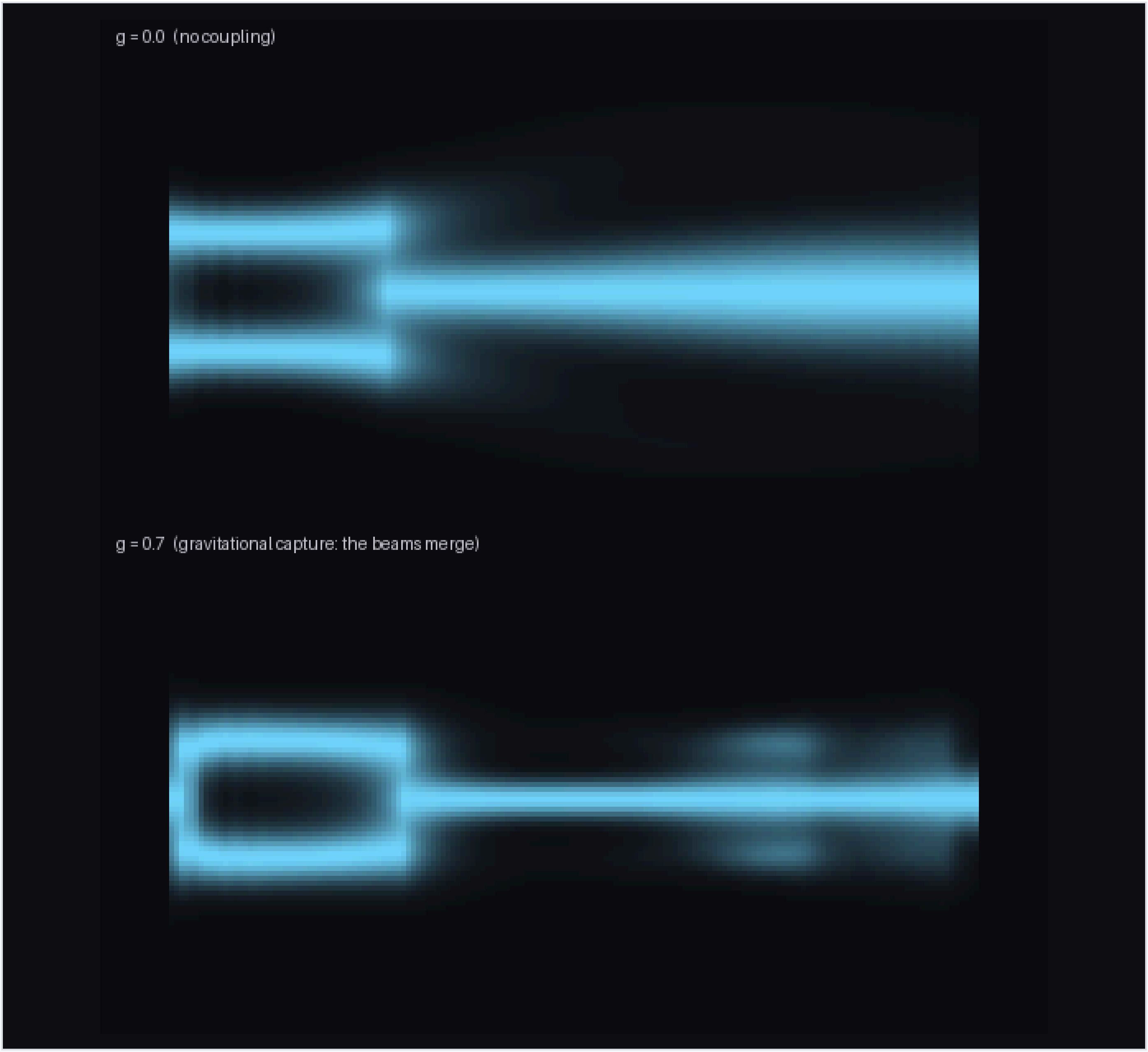

Figure 4. Self-gravity: long-exposure streaks of two packets. Top: no coupling, parallel flight. Bottom: energy-sourced potential — the beams fall together and merge (gravitational capture).

The metric becomes dynamical. Promoting the potential to a field with its own wave equation — on the same lattice, with the same stencil, at the same tick as matter (repository, part 18) — produces gravitational radiation and, with it, the two sharpest confrontations with data in this report. First, *the one-cone race*: a photon and a metric pulse emitted by one event cross a 450-cell track and arrive bit-identically, because a single substrate imposes a single dispersion relation — equality of c_{gw} and c_{γ} is an identity of the architecture, not a tuning. A deliberately built two-substrate engine (gravity updated on its own stencil) loses the same race by half the track. Observation has already performed this comparison: GW170817's gamma rays trailed the gravitational chirp by 1.7 s across ~40 Mpc, $|\Delta c/c| < 3\times10^{-15}$ — two-substrate designs are excluded by fourteen orders of magnitude, and one-substrate designs predict exactly zero, forever. `[Class A; passed]` Second, *the polarization test, which this toy's own variant fails*: a passing scalar-metric wave strains rulers along the propagation direction as strongly as across it (measured longitudinal/transverse response 0.94, correlation +0.93 — a breathing mode), whereas general relativity's tensor waves are transverse and quadrupolar, and LIGO–Virgo polarization analyses favor pure tensor. The scalar-metric variant of render gravity is therefore excluded by observation — measured here, conceded in advance by the honest accounting above. What survives is the tensor-metric program. `[Class A; the variant fails]`

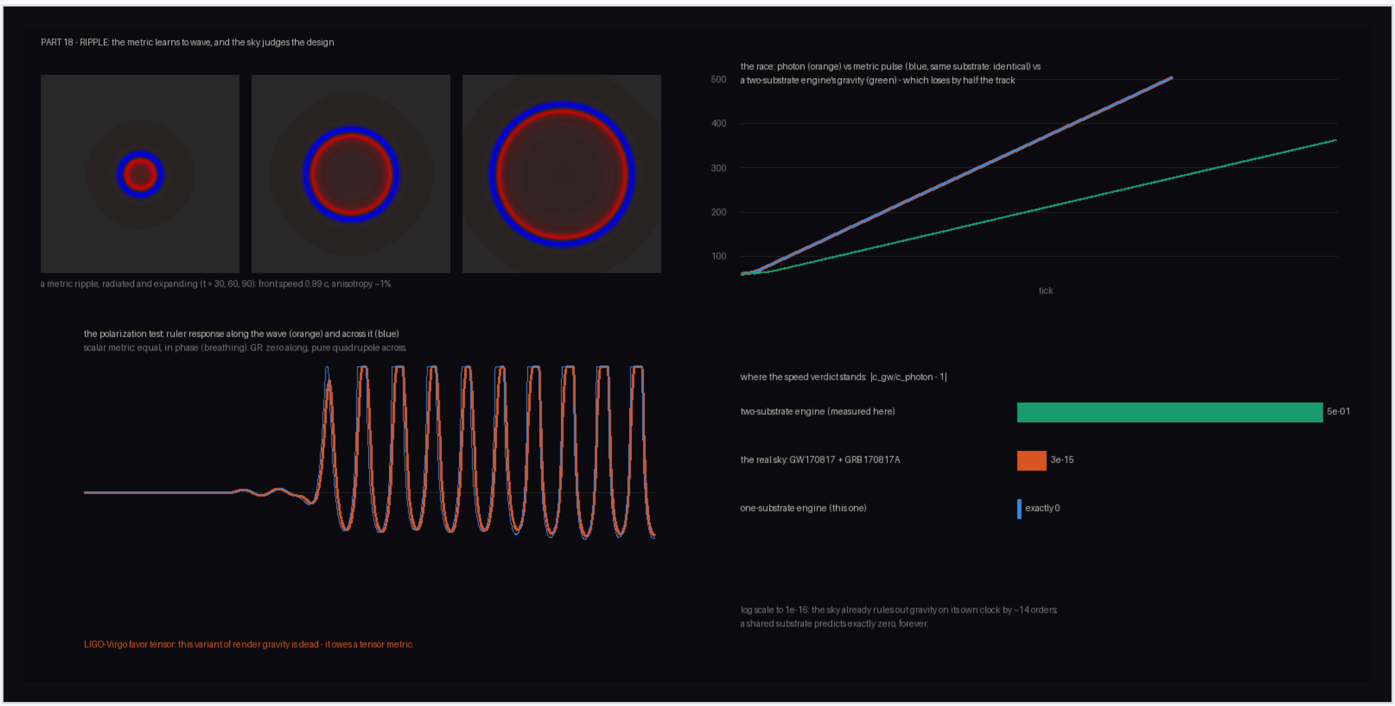

Figure 5. Gravitational-wave comparisons with observation. Top left: a radiated metric ripple (isotropic, front speed 0.99c). Top right: the race — photon and metric pulse overlapping exactly; the two-substrate engine's gravity losing by half the track. Bottom left: the breathing-mode strain (equal longitudinal and transverse response) that LIGO–Virgo excludes. Bottom right: the $|\Delta c/c|$ ladder.

Linearized tensor gravity. The surviving variant is then built (repository, part 21): linearized tensor gravity, h_{ij} on the lattice — in three spatial dimensions of necessity, since transverse-traceless waves do not exist in two (2+1 gravity carries no local degrees of freedom). Three measurements collect three of observation's verdicts. *The ring test*: an arrived wave strains a ring of test separations ($\delta L/L = \tfrac{1}{2}h_{ab}\hat{e}^a\hat{e}^b$, straight from the metric definition of length) in a pure $\cos 2\theta$ pattern — two polarizations 45° apart, mean response 0.007 of amplitude (traceless), longitudinal response zero — the pattern interferometers are built around, where the scalar of part 18 gave pure monopole. *The Birkhoff test*: with matter as an evolving field, whose stress is conserved because it obeys its own equation rather than by decree, a spherically breathing source leaves the strain channel silent to machine precision (exactly, by symmetry, as Birkhoff's theorem demands — its pulse arrives only in the strain-free trace channel), while two colliding packets — the binary analog — ring the detector at strain-per-trace efficiency 0.35. Monopoles cannot ring a gravitational-wave detector; that is why the sky's sources are binaries. *The Eddington test*: with slow matter calibrating the Newtonian deflection for the same well (each measurement minus its zero-gravity control, which drifts ~10 cells from diffraction alone), light on the full tensor metric bends **1.93×** the g00-only ("Newtonian light") amount — general relativity's factor is 2.00, the 1919 eclipse measured it, and Cassini pins $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$. Part 16's optical metric stands retro-diagnosed as Newtonian light. `[Class A; the surviving variant passes]`

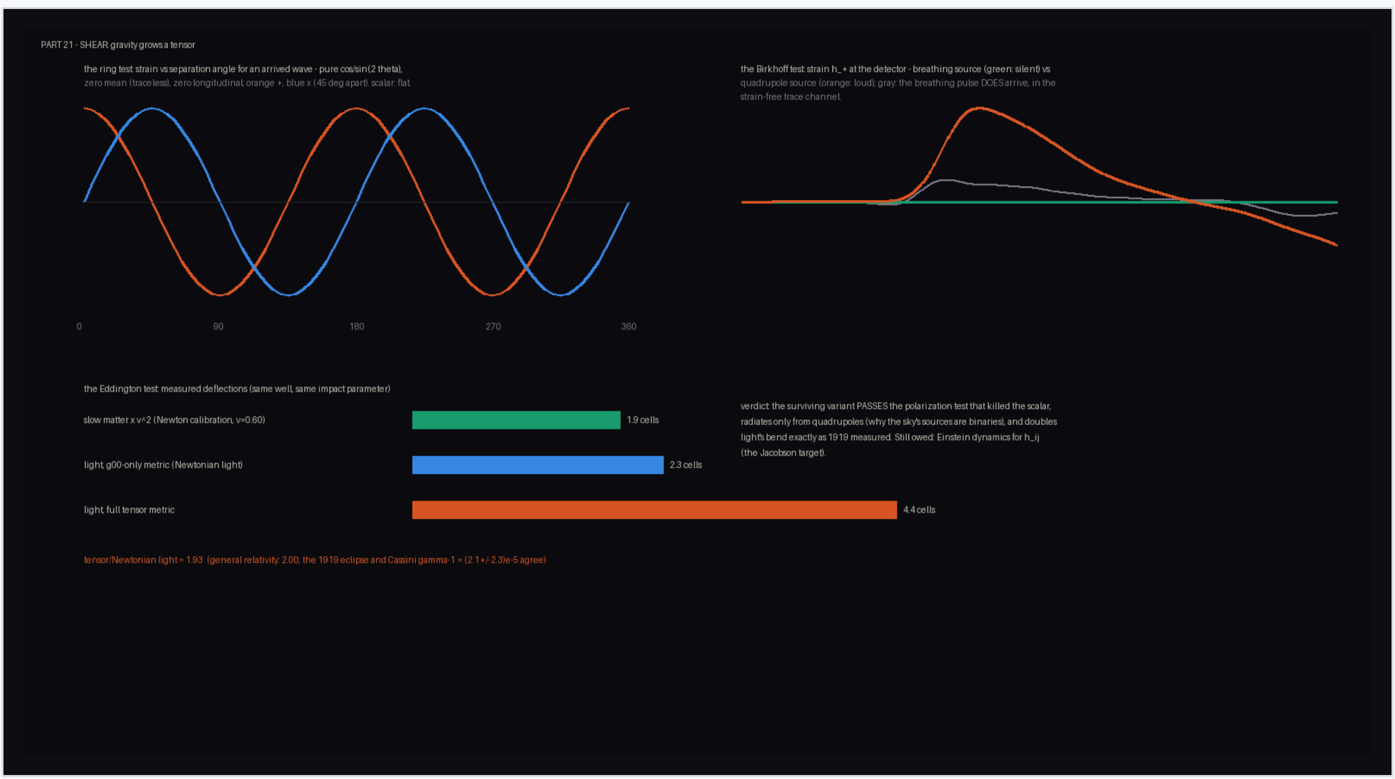

Figure 6. Tensor gravity, measured. Left: the ring test — strain versus separation angle, pure quadrupole, the + and × patterns 45° apart. Top right: the Birkhoff test — the colliding-packet (binary analog) source rings the strain channel; the breathing source arrives only in the strain-free trace channel. Bottom: the Eddington test — tensor light bends 1.93× the Newton-calibrated amount (GR: 2.00).

The entanglement first law. What remains owed after the tensor is the field equation itself, and Jacobson's program says it should not need postulating: gravity is the thermodynamics of entanglement — $\delta Q = T\delta S$ across local horizons — with the sharpening due to Faulkner, Guica, Hartman, Myers and Van Raamsdonk that the entanglement *first law*, holding for all regions at once, *is* the linearized Einstein equation. The first step of that program is now carried out on the Gaussian observatory of §5 (repository, part 22): a critical fermion chain on which every entropy is exact. Heated globally, the entropy change of every interval at every temperature collapses onto the parameter-free CFT curve $\frac{1}{3}\ln(\sinh x/x)$, $x = \pi\ell T/v$ (measured 1.020±0.026 across a 16× range of scales, Stefan–Boltzmann's $\pi/12$ measured en route); the small-x limb is the first law $\delta S = \delta\langle K\rangle$ itself, with the parabolic modular kernel — the inverse local Unruh temperature — carrying no fitted parameters (ratio 0.98±0.04), and the large-x bend is Bekenstein's bound $\delta\langle K\rangle \geq \delta S$, never once violated down to the measured instrument floor. The first law distinguishes heat from work: a pure particle changes a region's entropy only while it straddles the entangling cut (0.91 bits at the cut, 2×10^{-5} nats once fully inside, even as its modular energy grows without bound), a classically mixed packet contributes exactly its mixing entropy $h_2(\lambda)$ at any position (match 0.9999), and only thermal perturbations obey $\delta S = \delta\langle K\rangle$ at first order. The kernel can also be inverted: across a landscape of two warm regions, every sliding interval's δS is predicted at $r = 0.9997$ with no free parameters, and the inversion recovers the energy distribution from entropies alone ($r = 0.9996$). Finally, the measured small-x limb scales as $x^{1.98}$ (the exact curve's own slope over the fit window is 1.97; the pure limb is 2.00), and through the Ryu–Takayanagi dictionary — installed here, as the Born rule was in §7 — a parabolic kernel is the chord integral of a bulk response $h \propto z^2$, which is the *unique static solution of the linearized Einstein equations in AdS_3*; a bulk field of dimension Δ would respond as z^Δ. A mass gap destroys the z^2 behavior beyond the correlation length (the same probes fall to 0.00–0.68 of the curve), and hard localized excitations never produce it — a measured lesson from a failed first design, in which a point-source probe of the kernel failed exactly as Casini's relative-entropy form of the Bekenstein bound requires. The first law, and with it the geometric response, holds only in the thermal sector — which is precisely the δQ appearing in Jacobson's argument. `[Class A for the measurements; Class B for the CFT reproductions; the RT dictionary is an installed component]`

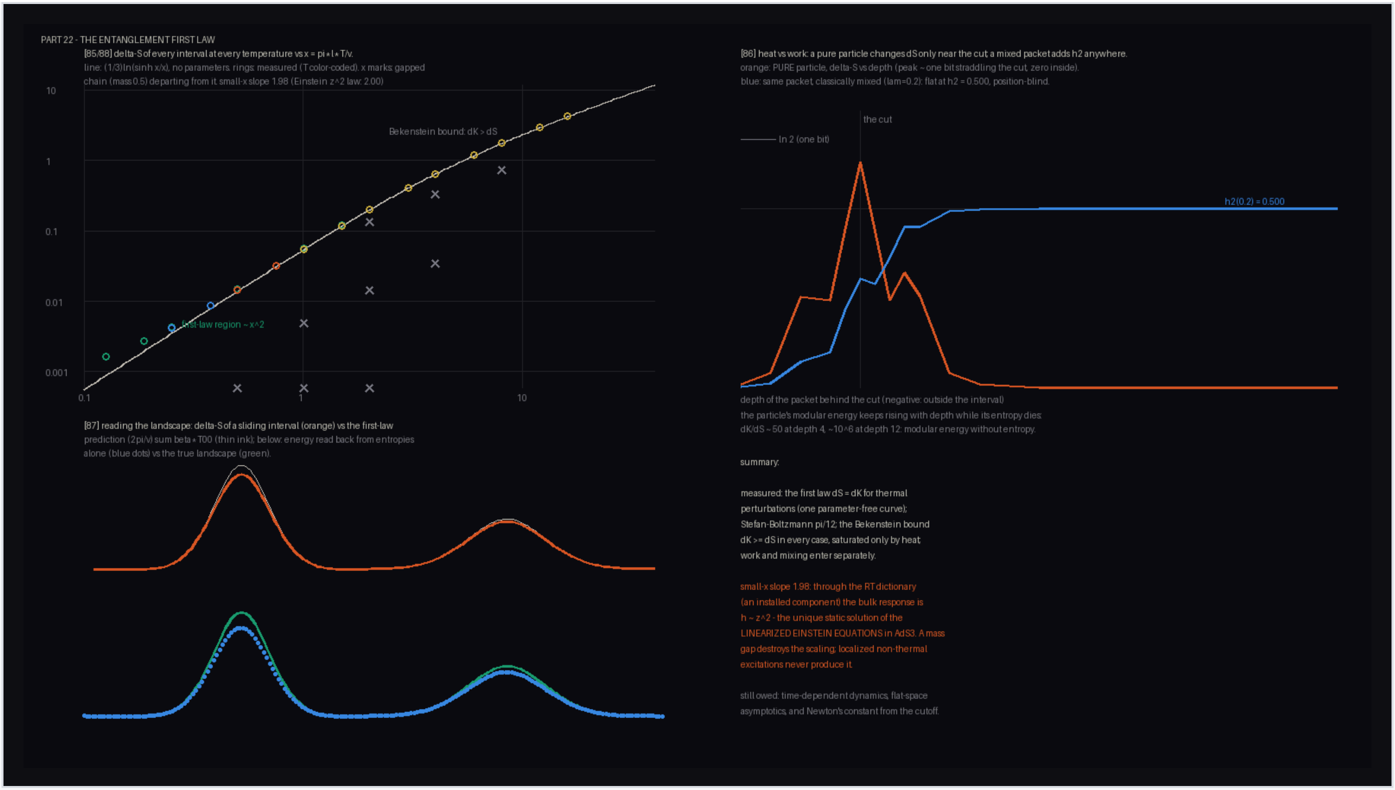

Figure 7. The entanglement first law, measured. Top left: δS of every interval at every temperature against x = πℓT/v, on the parameter-free CFT curve; × marks: a gapped chain departing from it. Top right: heat versus work — a pure particle changes the entropy only near the entangling cut (about one bit while straddling it), while the same packet, classically mixed, contributes exactly h_2 at any depth. Bottom left: two warm regions — the sliding-interval δS against the parameter-free kernel prediction, and the energy distribution recovered from entropies alone. The small-x slope, 1.98, corresponds through the RT dictionary to the z^2 profile of linearized Einstein gravity.

The first law in time. The measurement above is static; the time-dependent form is checked directly (repository, part 25). A warm region prepared on the same chain and released splits into two thermal pulses moving at the Fermi velocity — measured front speeds 2.00 for the energy and 2.01 for the entanglement of sliding intervals, against $v_F = 2$ — and the local first law holds at every time slice: the kernel prediction matches the measured δS at 1.030 ± 0.004 across the run, rising from 1.02 at interval half-width 8 to 1.10 at half-width 24 (the same interval-size correction as the static case), with total energy conserved to numerical precision and late slices excluded once the pulses dilute below the measurement floor. Through the Ryu–Takayanagi dictionary this says the AdS_3 metric response tracks the boundary stress tensor on each time slice. That is the correct behavior for three-dimensional gravity, which has no propagating degrees of freedom — the same fact that forced the tensor construction into 3+1 dimensions — so causality is carried by the stress tensor itself, visible as the light cone in the maps. The 3+1 version, where the metric propagates, is not delivered; the closing box says so. `[Class A]`

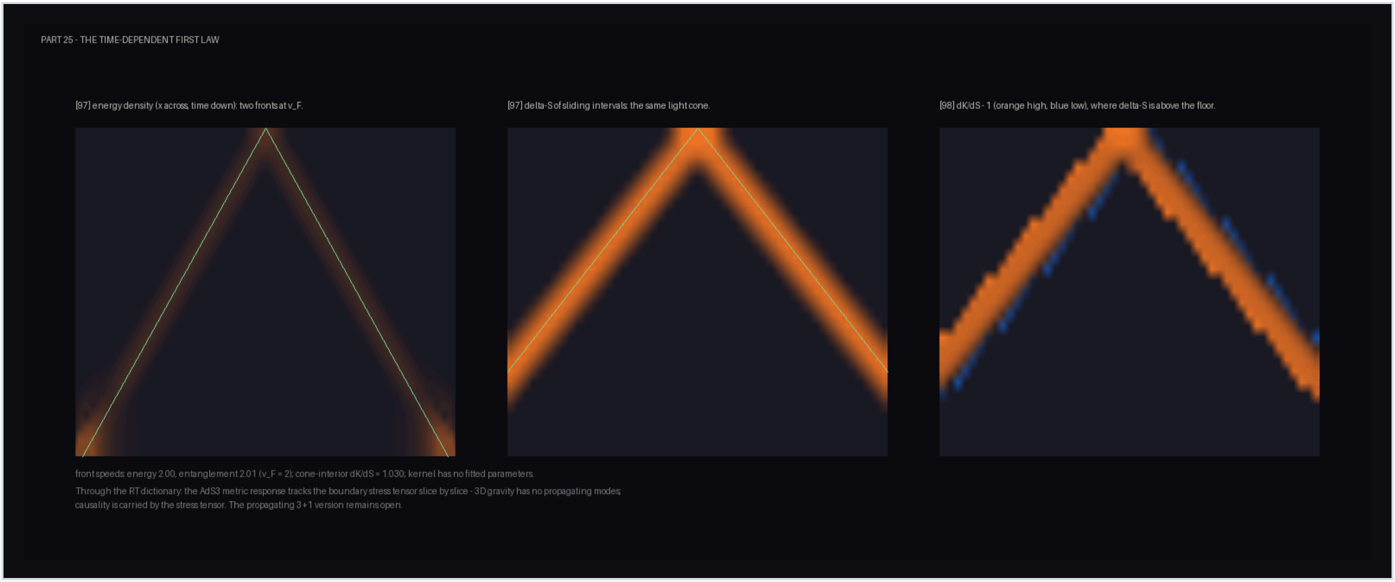

Figure 8. The time-dependent first law. Left: energy density of the released warm region (position across, time down) — two fronts at the Fermi velocity. Center: the entanglement of sliding intervals shows the same light cone. Right: the ratio of the kernel prediction to the measured δS, within a few percent of one wherever the signal is above the measurement floor.

Time dilation, horizons, and Hawking radiation. With the linearized field equation in hand, the natural next constructions are the objects it is known for (repository, part 23). Time dilation is measured directly, both kinds, with no fitted parameters: two identical cavity clocks at different depths of a well tick at the ratio $\sqrt{g_{00}}$ to four decimal places (0.5479 measured versus 0.5477 at $2|\Phi| = 0.7$), and a moving packet's internal phase clock — read at its own centroid — slows as $m\sqrt{(1-v^2)}$ out to $v = 0.73$, where the lattice's k^4 dispersion correction becomes visible. A hole is then built two ways, and the render distinguishes them sharply. The *frozen star* — a metric whose local light speed vanishes linearly at r_h — casts a shadow at the ray-traced critical impact parameter ($b_{crit}/r_h = 2.38$ on this profile; Schwarzschild's photon sphere gives 2.60), and an infalling packet's arrival times track the diverging metric integral $\int dr/c$ to 5% — the freeze — until the blueshifted wavelength reaches the lattice scale, where the substrate pins the packet and then *reflects* it, 99 ticks later: on a lattice with normal dispersion, a frozen star is a mirror with divergent delay, not a trap. A general-relativistic horizon is better modeled by the second construction: in Painlevé–Gullstrand coordinates space flows inward through a static frame, and the analog is a flow profile crossing $c = 1$ — Unruh's acoustic horizon. Evolving the chain's exact Gaussian state through the flow's switch-on, the horizon emits a steady Planck-shaped flux that was not put in by hand: measured temperature $0.94 \times \kappa/2\pi$ across two decades of occupation, against a curve computed from the flow profile rather than fitted. The correlation map resolves every emitted quantum's partner falling inside: the measured correlation ridge lies within 6 sites of the parameter-free predicted trajectory, with the partner remaining near the horizon before separating at $|v|-c$ — the same exponential peeling that makes the spectrum thermal. Meanwhile radiation–interior entanglement grows from 3.3 to 6.3 nats while the global state remains pure to ν_{min} = 0.5000: the radiation alone is thermal while the total state is pure, the same relationship between coarse-grained and exact descriptions as in §8. Lattice dispersion enters both constructions, oppositely: it is why the frozen star reflects, and it is why the acoustic horizon radiates *steadily* (dispersion supplies the horizon with fresh modes — Unruh's analog-gravity observation) while the spectrum stays Planckian regardless (Corley–Jacobson). That the Hawking temperature does not depend on the substrate's microphysics is a robustness statement this program in particular needs. `[Class A for the measurements; Class B for the reproduced physics (Hawking 1974; Unruh 1981)]`

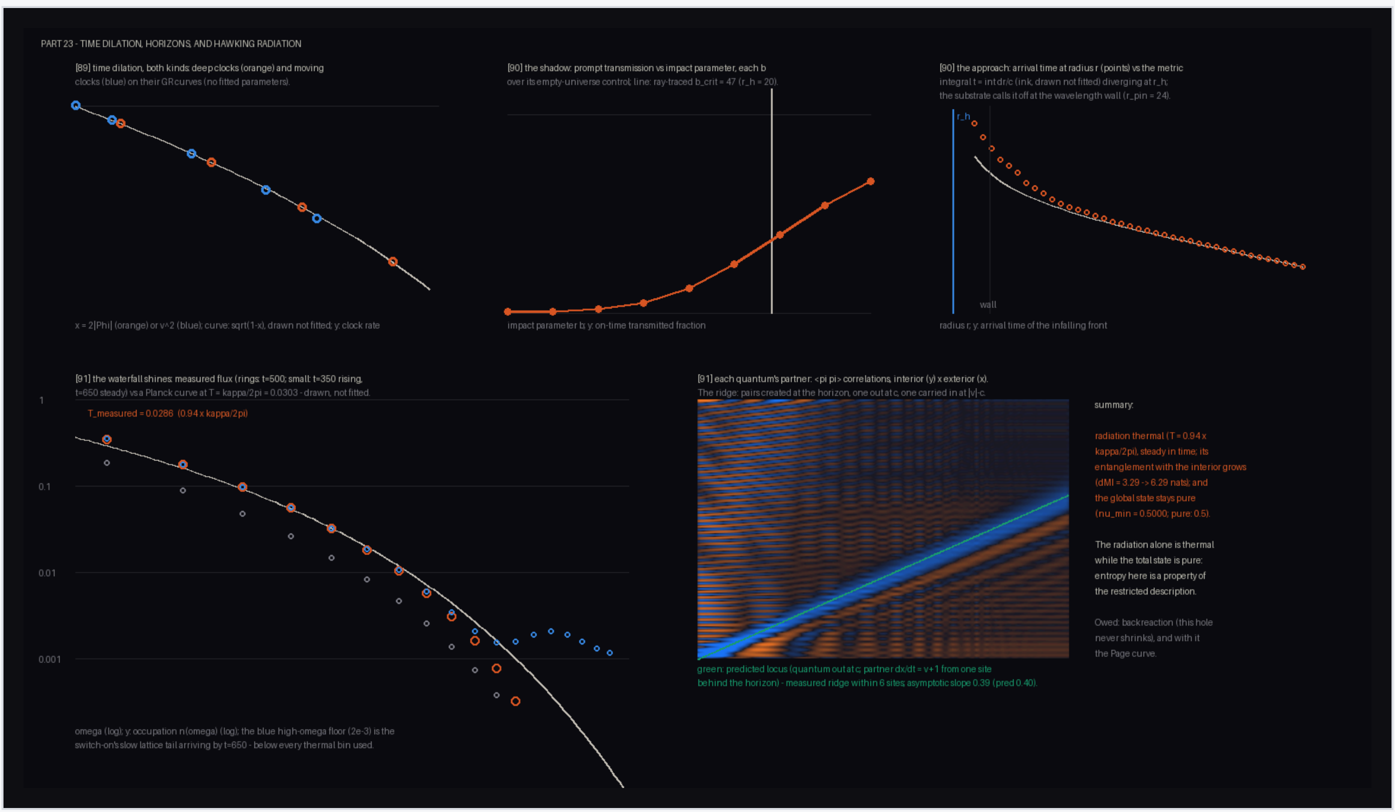

Figure 9. Horizons, measured. Top: both time dilations on their GR curves (no fitted parameters); the photon shadow against the ray-traced critical impact parameter; infalling light following the diverging metric integral until the lattice-scale cutoff. Bottom: the steady Hawking spectrum against a Planck curve at κ/2π (computed, not fitted), and the partner correlations — each emitted quantum's twin falling inside along the predicted trajectory.

Evaporation and the Page curve. The acoustic horizon above radiates but does not shrink, so the entropy of its radiation can only grow — Hawking's calculation, and his information paradox. The final construction lets it evaporate (repository, part 24). The matter is the fermion chain of §6's first-law measurement, with the flow supplied by an imaginary second-neighbor hopping (a one-dimensional form of a type-II Weyl horizon); every state is then a Slater determinant, evolution is a sequence of exactly unitary two-site rotations, and the fine-grained entropy of the radiation is computable exactly at all times — which is what makes the experiment possible: the island formula can be checked against the true answer. First, statistics: the same surface gravity that gave bosons a Planck spectrum gives fermions a Fermi–Dirac spectrum at the same $\kappa/2\pi$ (measured ratio 0.98 over two decades, particles and holes emitted symmetrically) — temperature depends only on the geometry, statistics only on the matter. A hole whose flow is just 20% past critical also stops radiating above about $3T_H$: the dispersive (trans-Planckian) cutoff of Corley–Jacobson, measured. Then the horizon retreats under an energy ledger: the hole is assigned a budget $M = \rho x_h$, the outgoing flux through a fixed surface is measured during the run, and the position follows $dM/dt = -F(t)$ — so the lifetime is measured, not scheduled. The retreat stays faster than the stored interior partners drift (each is overtaken and released), accelerates as its own emission adds to the flux, and ends at $t = 492$. Three curves are computed for the entropy of the radiation, defined as everything outside the hole. The *exact* entropy rises while the horizon radiates, turns over at $t = 359$, and returns to zero when the hole is gone: unitarity, verified directly (the global state remains pure to 10^{-12}). The *extrapolation at the measured early emission rate* climbs to 7.9 nats and never turns over: Hawking's curve, and the gap between the two is the information paradox. The *island formula* — minimum over islands I of S(radiation ∪ I) plus an area term — uses one measured constant: $\mu = 0.36$ nats per cut, the chain's vacuum entanglement across a single cut, which is precisely the induced-gravity identification of $1/4G$ with the UV entanglement density (Susskind–Uglum). It crosses the extrapolation at $t = 120$ and stays within 0.51 nats of the exact curve on average, with the optimal island tracking the stored partner quanta. Why the formula works is visible in the toy: the island contains the partners, so radiation-plus-island is nearly pure, and the remaining cost is the area term. Two honest differences from the gravitational case are stated in the output: in one dimension the horizon is a single cut, so the island branch is roughly constant rather than falling with a shrinking area;

and this chain has no replica wormholes, so the exact curve turns over when the partners are physically released rather than at the formula's crossing. `[Class A for the measurements; Class B for the reproduced physics (Hawking 1974; Page 1993); the island rule and the retreat schedule are installed components]`

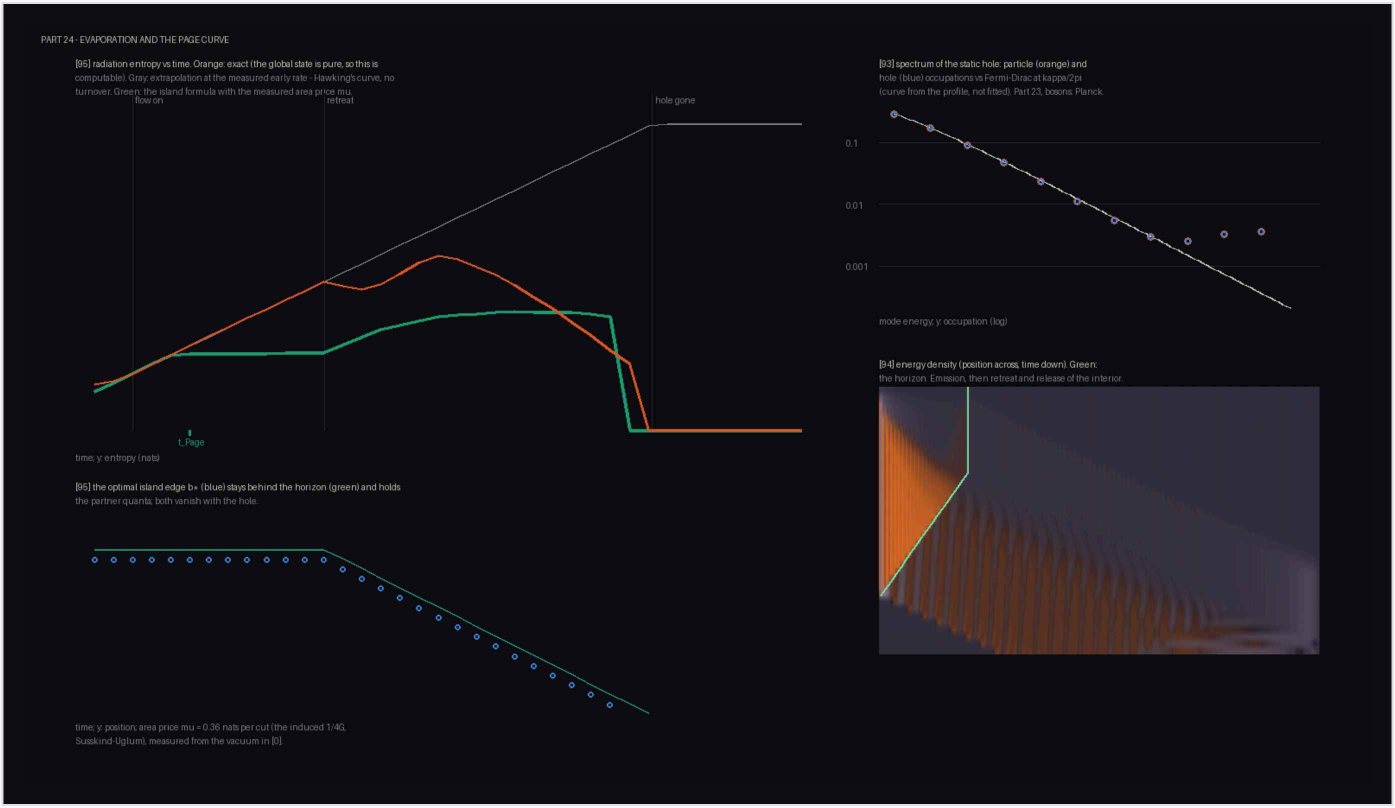

Figure 10. Evaporation. Left: the three entropy curves — exact (orange: rise, turnover, return to zero), the never-turning extrapolation (gray), and the island formula with the measured area term (green) — with the optimal island edge tracking the horizon below. Right: the Fermi–Dirac spectrum of the deep static hole, and the energy density over time showing emission, retreat, and the release of the interior.

WHAT THIS SECTION DOES NOT DELIVER

Three things remain assumed rather than derived. (1) Propagation: the first law, static and time-dependent, is established in AdS_3, where the metric response has no dynamics of its own; the 3+1 version — a time-dependent first law sourcing the propagating h_{ij} of the wave experiments — is not constructed, and neither is a flat-space form of the correspondence. (2) Backreaction: the evaporating horizon's position is driven by the measured flux, but through a prescribed energy-per-length constant, a prescribed profile shape, and a rate cap in the final runaway; no first-principles energy budget for the flow itself is given. (3) The island rule is imported: real gravity derives it from replica wormholes, and nothing in these models establishes that identification independently.

7 The quantum render layer

Wave–particle duality, in this architecture, is a type distinction rather than a paradox: the wave is the engine object; the particle is the render event. A double slit run in the wave engine shows one field passing through both slits (five fringe maxima; spacing 14.5 cells against the textbook $\lambda L/d = 12.2$ with near-field corrections), while a Born-rule click sampler at the screen assembles the same fringes dot by dot, Tonomura-style. The Born rule is installed here (as in §3); the demonstration is architectural. `[Class B, installation disclosed]`

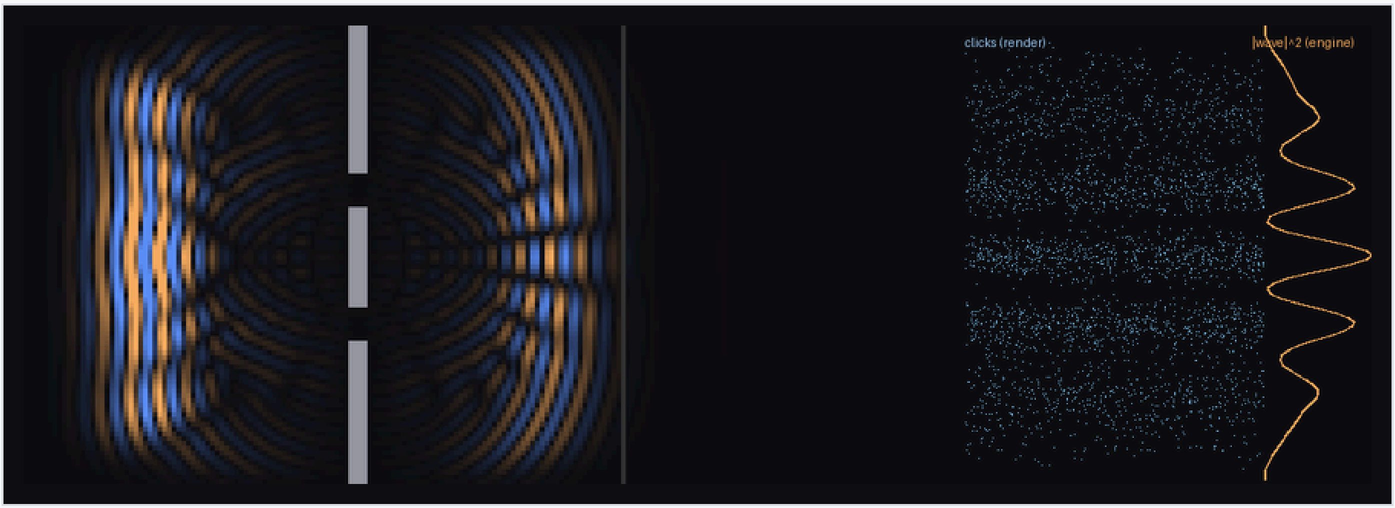

Figure 11. Duality as architecture. Left: the engine's wave interferes through both slits. Right: the renderer's discrete clicks reproduce $|\text{wave}|^2$ statistically.

What no classical render layer can do is also quantified: with both slits open, arrivals at the deepest minimum fall to **4.8% of the sum of the single-slit rates** — below each single slit alone at 10 of 36 central positions. Any engine in which each particle takes one slit, without slit-to-slit action at a distance, is bounded below by P_1+P_2 everywhere. Amplitudes cancel; probabilities cannot. The classical cost of faking it is likewise measured: statevector simulation cost multiplies by 2.14 per added qubit (10^{19} years by 100 qubits). `[Class A/B]`

Measurement completes the picture. A which-path detector (a phase-kicking cell in one slit) degrades fringe visibility 0.90 → 0.49 as its coupling grows, tracking a phase-average prediction; and the washed-out ensemble, *re-sorted by the detector's record*, recovers visibility 0.89–0.93 per bin with fringes shifted per recorded kick. Nothing collapses; coherence moves into correlation, and conditioning on the record retrieves it. `[Class A/B]`

The architecture also gives the interpretation debate a compact form: *one engine state, many render threads*. "Branches" are decoherence-separated slices of a single engine state, exactly as separable as the records that define them — the eraser above is a small branch reunification. The live interpretations then sort by what the engine keeps. Everett: everything (pure unitarity — the reading most congenial to this program's inviolable reversibility axiom, and the one whose Born rule the envariance argument of §4 serves). Objective collapse (GRW; Penrose): a pruning, garbage-collecting engine — natural for a resource-bounded substrate (§11), and testable, since pruning predicts small unitarity violations of the kind now being bounded by large-molecule interferometry. Bohmian mechanics: everything, plus a pointer marking the occupied thread. A fault-tolerant quantum computer is a machine that forces the engine to keep branches coherent at scale, so the quantum-computing fork of §11 doubles as an experimental referendum on the engine's bookkeeping. `[Class C]`

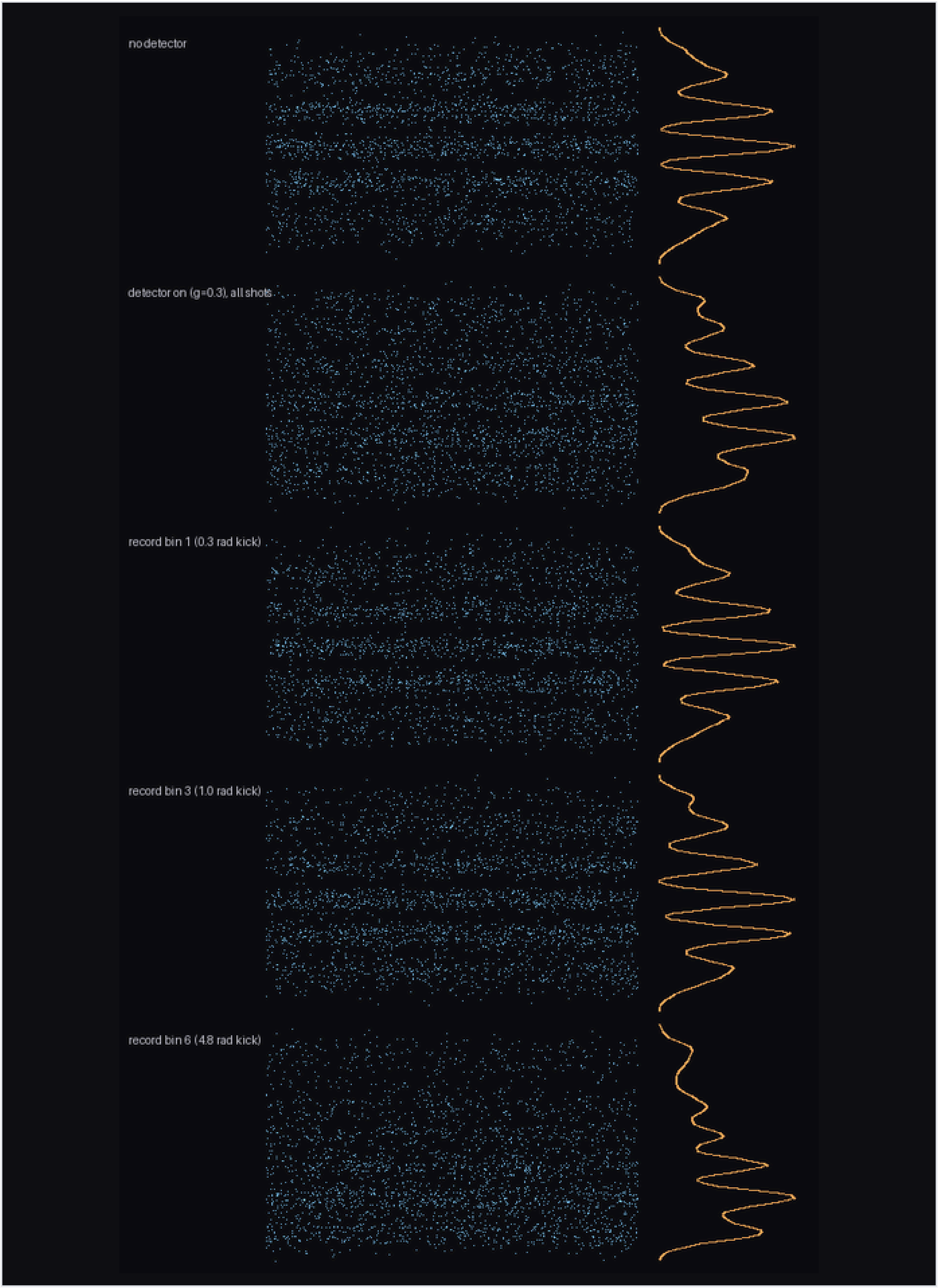
no detector
detector on (g=0.3), all shots
record bin 1 (0.3 rad kick)
record bin 3 (1.0 rad kick)
record bin 6 (4.8 rad kick)

Figure 12. Measurement without collapse. Top to bottom: no detector (fringes); detector on, all shots (washed out); the same shots sorted by the detector's record (fringes return, shifted per record).

8 The arrow of time, and observers as dynamical patterns

The engine is a bijection, so the arrow cannot live in the laws. Coarse-graining the render into blocks and counting Boltzmann microstates: a 2,086-particle blob's entropy climbs 3,831 → ~7,160 bits; exact reversal marches it back to 3,831 with bit-perfect recovery of the microstate; flipping one cell in the gas before reversing corrupts 21% of the torus and the past is gone. Where the flip lands matters completely: in vacuum, the perturbation curls into a stationary bound state and the past survives minus one cell. **Chaos in this universe is matter-borne**: histories are destroyed by interaction and preserved by isolation — the entropy arrow and decoherence as one phenomenon. Random states of equal matter count all sit at $S = 8{,}126 \pm 16$; the initial blob is one microstate in 2^{4295}. `[Class A]`

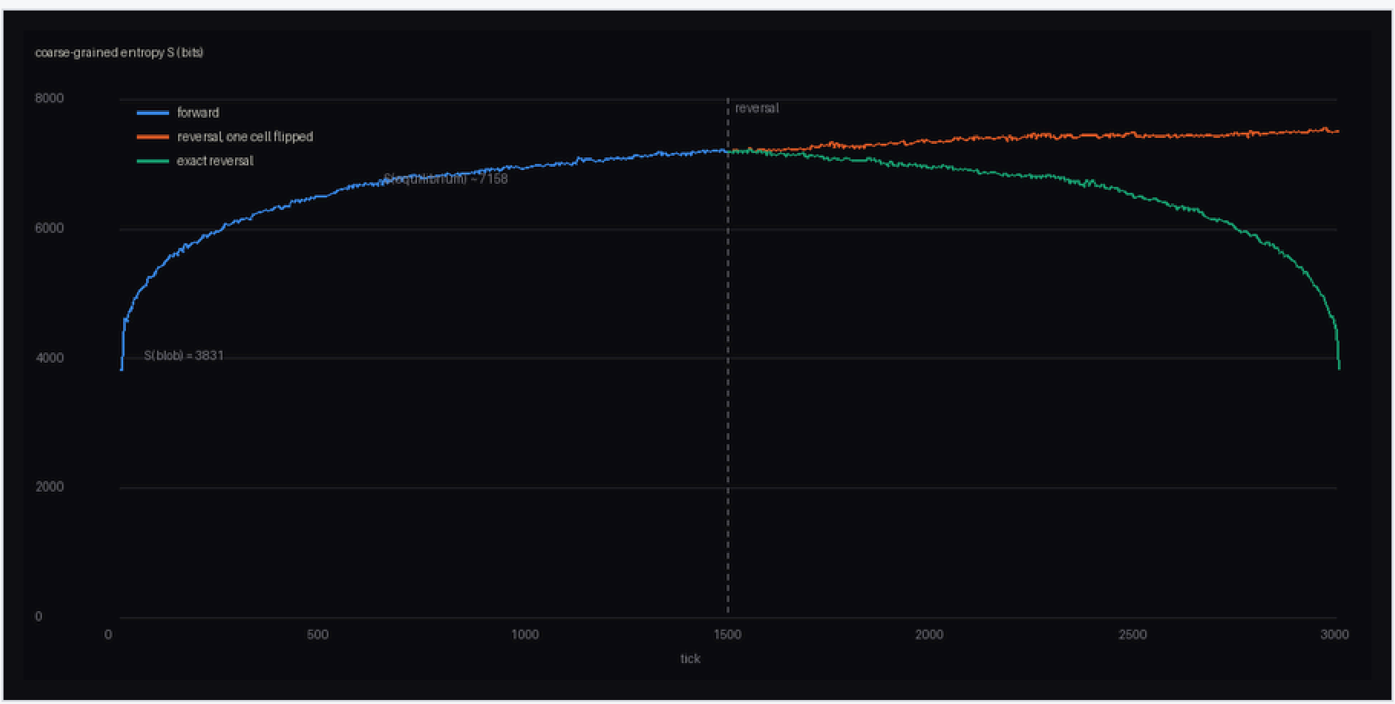

Figure 13. The arrow is not in the laws. Entropy rises (blue), retraces exactly under reversal (green), and clings to equilibrium when one cell was flipped first (orange).

Observers require no extra physics. Colliding a glider with each of 456 bound states across impact parameters, **21% of 3,192 collisions leave a stable, isolated, changed local structure** — a one-bit record written by a physical event, intact 400+ ticks in isolation. Detectors are weather. `[Class A]`

9 Hiding the lattice

The loudest structural objection — grids have preferred directions; Lorentz symmetry is exact — has a known escape (causal sets), demonstrated here quantitatively. A lattice's light-speed anisotropy is systematic (20% at wavelength 4, zero variance) and a boosted spacetime lattice shifts its nearest-neighbor statistics by 29% with zero variance: every observer measures an absolute velocity. A Poisson sprinkle's anisotropy is statistical (direction-random, inside its own noise) and its boosted statistics shift by 1%: an area-preserving map of a Poisson process is the same process. Discreteness without grain costs noise — waves on naive random graphs scatter — but noise averages away with scale separation, and nature's is 10^{20}. `[Class A/B]`

10 The axiom collider

The program's method, made explicit: physics as constraint satisfaction over "obvious" axioms. Fifteen collisions are catalogued ([`COLLIDER.md`]() in the repository), each Compatible, Converging, Incompatible, or Open, with authority cited — a named no-go theorem or an in-repo measurement (12 of 15). The no-go theorems are the fossil record of collisions past; the black-hole information paradox is one in progress. Two entries deserve emphasis. **Weinberg–Witten (1980)**: a massless spin-2 graviton cannot be composite in the same spacetime — emergent gravity is consistent only if the space it curves is itself part of the emergent layer. The theorem very nearly demands the render architecture. `[Class B/C]` And the render frame *ranks* the axioms in the live crash: engine-level reversibility is inviolable, render-level locality is negotiable — the direction the replica-wormhole results took in 2019. `[Class C, post-dictive]`

11 Predictions: the seed and its fingerprints

The theory's explanatory target is the Past Hypothesis — the unexplained, preposterously low entropy of the initial state. The render frame's answer: initializations are written, written things are compressible, and compressible states are low-entropy states (our initial blob: fifty characters of code, one microstate in

2^{4295}). `[Class C]` This fuses with a prediction: if the seed's generator is simple, the primordial field is pseudorandom, not random, and carries fingerprints.

Two measured laws discipline the prediction. **The sensitivity ceiling:** a blind detector battery (spectral, autocorrelation, pair-structure, GF(2) parity, compression) catches tiled seeds, 16-bit LCGs and 31-bit LFSRs with 2^{12}–2^{14} samples, but a 40-byte 32-bit LCG is already invisible at 2^{20} — detectability dies while simplicity remains. **The survival law:** under 600 steps of chaotic evolution, statistical pseudorandom structure launders completely (calibrated differential test: $z = 5.4$ against a null band of 4.9–5.9), while an *exact seed symmetry* survives at $z \approx 200$ — translation-equivariant dynamics cannot break an exact symmetry of the initial state. `[Class A]`

Therefore: the only robustly observable fingerprint class is **seed symmetry and long-range alignment, in the earliest accessible layer** — which, for the real universe, is the CMB. The observed low-ℓ anomalies (quadrupole–octupole alignment ~9°, hemispherical power asymmetry $A \approx 0.07$, low-ℓ odd-parity excess) are statistics of exactly this class, individually ~2–3σ, unexplained by inflation and conventionally dismissed as flukes.

A spherical detector battery was built to make this quantitative: an import-time-validated spherical-harmonic transform (Gauss–Legendre × FFT, round-trip $< 10^{-10}$), the de Oliveira-Costa preferred axis computed via the angular-momentum inertia tensor, hemispherical asymmetry, parity ratio, and spectral spikes, all calibrated on 200 Monte-Carlo isotropic skies (false positives 1/60 at $p<0.01$). Injecting the three anomalies at their published amplitudes into a single $\ell \le 24$ sky: individually $p = 0.047$, 0.065, 0.045 (the modulation is lost to cosmic variance at this ℓ-range), **jointly $p = 4.3\times10^{-3}$** (≈2.8σ) under Fisher combination. Under the hypothesis that predicts them together, the anomalies are one moderately significant signature, not three dismissible flukes. Sharper still: an exact seed symmetry is not a statistic but a *spectral selection rule* — a mirror-symmetric seed yields odd-(ℓ+m) power fraction 0.000 against an isotropic 0.5. A dedicated selection-rule search on Planck maps is the stated experiment; its first sweep follows below. Because the temperature anomalies were identified *a posteriori*, the joint statistic above measures coherence under a stated hypothesis, not discovery significance; the clean test is therefore registered here in advance for statistically independent probes of the same modes — E-mode polarization foremost — where a common physical origin predicts the anomalies recur and flukes predict they vanish. `[Class A for the instrument; Class C for the cosmological inference]`

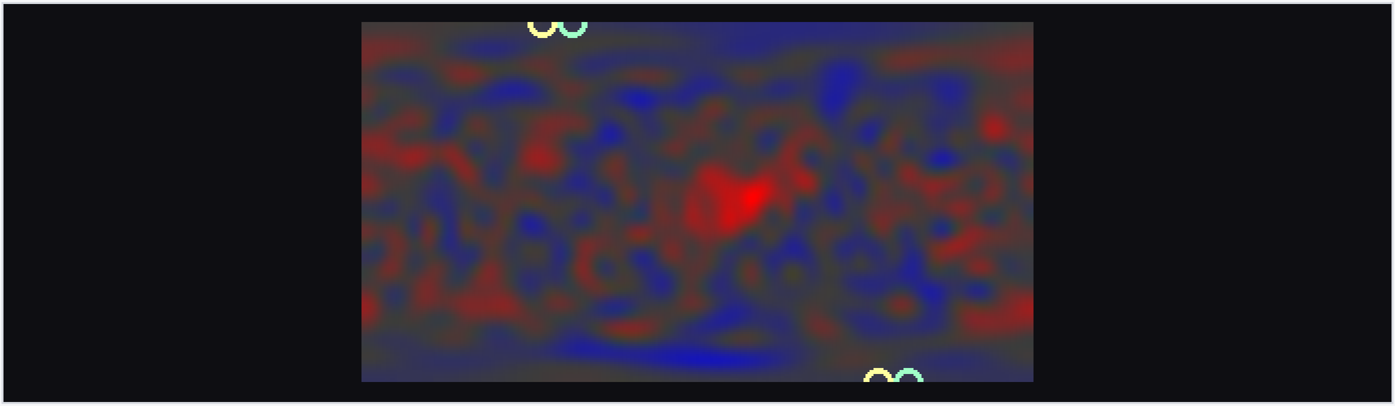

Figure 14. A synthetic sky carrying the three observed anomalies at published amplitudes; rings mark the aligned quadrupole and octupole axes.

First light. The battery has now been pointed at the actual sky (repository, part 19). Chain of custody: the four Planck 2018 component-separated temperature maps (SMICA, Commander, NILC, SEVEM; inpainted) and the WMAP nine-year ILC are read from raw FITS bytes by a hand-rolled, import-validated HEALPix reader — no external astronomy libraries — analyzed to $a_{\ell m}$, and pushed through the identical statistics against 10,000 Gaussian ΛCDM skies drawn from the published best-fit spectrum (URL and SHA-256 provenance of every byte stream committed with the reduced data). The instrument validates on the sky itself: the two missions agree multipole by multipole (mean $r = 0.98$ over $\ell \leq 24$), the four Planck pipelines pairwise at $r = 0.99$, and the from-raw-bytes spectrum reproduces the published low-ℓ points, including the low quadrupole ($D_2 = 198$–$213\ \mu K^2$ across Planck methods, against ΛCDM's 1017). The literature's anomalies then come out with calibrated p-values: quadrupole–octupole alignment 8.1° ($p = 0.010$; axes at Galactic (240°, 55°) and (239°, 63°), where the literature puts them, 16–20° from the CMB dipole axis), hemispheric asymmetry $p = 0.056$, odd-parity excess $p \approx 0.05$ — individually the familiar ~2σ curiosities. Jointly: **p = 0.003–0.013 across the four Planck maps** (2.5–3.0σ; WMAP's ILC, which carries larger residuals, gives $2{\times}10^{-4}$), matching the forecast from injections at published amplitudes — an end-to-end consistency check of instrument, nulls, and the literature's numbers in one inspectable pipeline. The mirror selection-rule scan — 240 axes, exact Wigner rotations, identical treatment for data and nulls — finds no rule: minimum odd-$(\ell+m)$ fraction 0.31–0.33, $p \approx 0.7$. That is the honest outcome for the sharpest test: *a bound, not a detection* — an exactly mirror-symmetric seed is excluded at these scales, and approximate symmetry is bounded by the scan floor. One curiosity is registered without a significance claim: the fraction-minimizing axis lands 5° from the CMB dipole in four of the five maps, the direction Planck's own mirror-parity tests flagged. The caveats of the preceding paragraph stand in full: statistics selected a posteriori by the field, temperature only, and the registered E-mode test remains the clean discriminator. `[Class A for the measurements; Class C for the cosmological inference]`

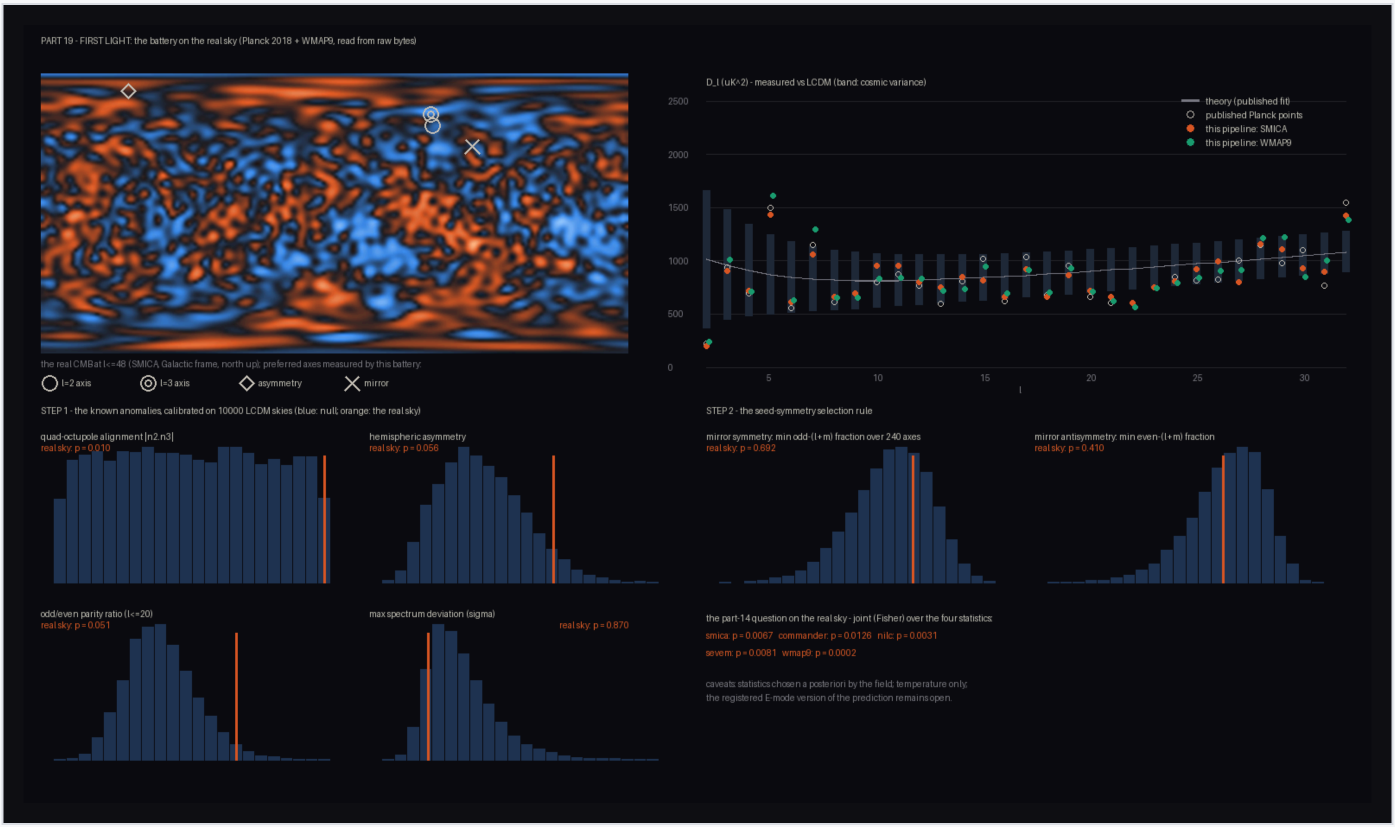

Figure 15. First light: the battery on the real sky. Left: the measured CMB (SMICA, $\ell \leq 48$) with the battery's preferred axes — the two rings nearly touching are the aligned quadrupole and octupole. Right: the from-raw-bytes spectrum against ΛCDM (the low quadrupole at far left). Below: each statistic against its 10,000-sky null (orange: the real sky).

The echo. Registration pays out immediately (repository, part 20). E-mode polarization at large angles — reionization re-scattering of the same super-horizon modes — is a partially independent second draw from the primordial field: seed features must recur there at the registered axes, flukes must not. A hand-rolled spin-2 transform (Wigner d-functions by recursion, seeds and signs pinned against the angular-momentum matrix exponential, pixel geometry pinned against reference HEALPix values) reads E/B from the raw Stokes bytes and calibrates absolutely: TE amplitude 0.97–1.00 against the ΛCDM prediction across all four Planck pipelines, TT at 1.000. It also catches a canary worth registering for others: *Planck's polarization* *inpainting, which touches 4% of sky, removes a third of the large-scale TE signal* (amplitude 0.68 on inpainted maps, uniformly across methods) — the battery therefore runs on the un-inpainted maps, with the inpainted variant as cross-check. The test is careful about the trap that T and E are correlated (a fluke partially echoes into E through C_ℓ^{TE}): the null ensemble draws E *conditional on the real temperature sky*, and the headline statistic is the T-subtracted residual. Five statistics at the registered axes — mirror fraction, both alignment angles, the asymmetry direction, parity — find **no echo**: joint p = 0.22–0.76 across methods. The power analysis says this outcome was forced either way: with the measured noise floor (taken from B, which cosmology leaves empty at these scales; signal-to-noise per mode falls from 1.4 at ℓ=2 to 0.05 by ℓ=16) the test would detect a genuine temperature-amplitude echo only **32%** of the time, versus **98%** at cosmic-variance-limited (LiteBIRD-class) sensitivity. The registered prediction is thus neither confirmed nor damaged — it now carries a measured sensitivity requirement, and a date: the next generation of polarization data decides it. `[Class A for the measurements and the power calculation; Class C for the cosmological inference]`

Figure 16. The echo test. Top left: the noise reality — measured EE (orange) sits on the B-mode floor (blue) rather than the ΛCDM E signal (gray) beyond ℓ≈8. Top right: the inpainting canary in TE. Bottom: the registered-axis battery (no echo) and the measured power of the test — 32% at today's noise, 98% at LiteBIRD-class.

What Earth can see. Three walls stand between an observer here and the seed, and they are of different kinds. The first is dynamical and already measured (§11's survival law): chaotic evolution launders the seed's *content* completely — detectability of a generator dies at ~40 bytes — leaving only its symmetry class observable. The second is informational and permanent: the signature lives at the largest angular scales, and the sky at $\ell \leq 24$ contains roughly six hundred numbers, total, for any instrument that will ever exist — cosmic variance is not noise but the fact of a single realization, which is why the temperature signature saturates near 3σ and why Planck's low-ℓ temperature data cannot be improved upon. The only way past it is new *channels* onto the same modes: E-mode polarization (the echo above, worth roughly a doubling once noise-free), CMB lensing, and ultimately dark-ages 21-cm tomography, which would replace our single two-dimensional slice of the primordial field with a three-dimensional volume carrying orders of magnitude more large-scale modes — the long-run arbiter of this program. The third wall is positional: the CMB sphere is centered on us, and a different observer sees a different slice. Global seed symmetries — the class the survival law selects — project onto every observer's sphere, which is what makes them testable at all; signatures tied to special *locations* in the seed could be arbitrarily diluted in our patch (one known partial escape exists in principle: the polarization of cluster-scattered CMB light encodes other observers' skies, a remote sampling of cosmic variance). The budget accounting, then: temperature has spent its share and sits at 2.5–3σ in the predicted direction; polarization's share is real and unspent; the decisive information lies in channels not yet opened. `[Class A for the measured walls; Class C for the outlook]`

A second observational fork runs independently, and is properly three-branched. If the engine is quantum, fault-tolerant quantum computation scales and efficiently *verifiable* advantage (Shor; Mahadev's 2018 classical-verifier protocol under LWE) holds up permanently. If the engine is classical and polynomial, then either quantum computers hit an engineering-independent ceiling, or classical computation secretly matches quantum (BQP ≈ P — unproven either way, and the repeated classical spoofing of "supremacy" claims keeps this branch honest); in the latter case a classical engine keeps pace only if the relevant cryptography is breakable in principle. A memory-bounded variant even yields a number: a classical engine that brute-forces amplitudes for entangled subsystems within the holographic budget (10^{122} bits; consistent with the area law of §5) supports faithful quantum computation only to

$n \approx \log_2(10^{122}) \approx 400$ logical qubits — beyond every current machine, inside fault-tolerance roadmaps. The bound is on the largest *jointly entangled block*, not on qubit inventory: independent devices cost the engine the sum of their exponentials ($2^a + 2^b$, never 2^{a+b} — only entanglement fuses ledgers), decoherence continually splinters blocks back into cheap factors, and nature's own entanglement is short-ranged area-law structure a bounded engine simulates in polynomial time; a render-aware simulator (repository, part 15) measures all three effects, including a maximally entangled 22-qubit state costing 248 bytes in stabilizer form until a single non-Clifford gate multiplies the memory cost 270,600-fold — magic, not entanglement, is the expensive resource. Sampling-style demonstrations cannot decide the fork (their verification is itself exponentially hard, so a classical engine could serve pseudo-samples indefinitely); verifiable advantage sustained in one coherent block at the few-hundred-logical-qubit scale can. The measured wall of §7 (×2.14 per qubit) sets the classical price; the experiment is industrially funded. `[Class C, falsifiable per branch]`

The scorecard

A hypothesis earns its keep by what could have killed it. The table collects every confrontation between this architecture and data — in the toys and in the sky — including the two variants observation has already executed.

PREDICTION	TEST	VERDICT
One causal cone ($c_{gw} = c_\gamma$)	toy: bit-identical race (part 18); sky: GW170817 + GRB 170817A	**passed** ($< 3\times10^{-15}$; one-substrate predicts exactly 0)
GW polarization — scalar-metric variant	toy: longitudinal/transverse = 0.94 (part 18); sky: LIGO–Virgo favor tensor	**variant excluded** → tensor metric required — and built: the tensor variant's ring response is quadrupolar, transverse, traceless (part 21)
Monopole silence (Birkhoff)	toy: breathing source strain = machine zero; colliding-packet quadrupole rings at 0.35 (part 21); sky: GW sources are binaries	consistent — the tensor variant radiates only from quadrupoles
Light bending factor 2 ($\gamma = 1$)	toy: tensor/Newtonian light = 1.93 (part 21); sky: 1919 eclipse; Cassini $\gamma-1 = (2.1\pm2.3)\times10^{-5}$	**passed** by the tensor variant; g_{00}-only light excluded
No preferred frame — lattice-substrate	toy: 29% boosted-frame shift vs sprinkle 1% (part 10); sky: clock-comparison &	**variant excluded**; sprinkle-class survives (costs noise, not

variant	GRB dispersion bounds	anisotropy)
Equivalence principle — medium-coupled variant	toy: chromatic bend/opacity (part 6) vs achromatic metric coupling + Eötvös pass (part 16); sky: MICROSCOPE 10^{-15}	**medium variant excluded**; metric coupling passes
Area-law entanglement (holographic data layout)	toy: S ∝ perimeter, r = 1.0000 (part 10)	consistent (Class B)
Joint low-ℓ CMB signature (common seed origin)	real sky, four Planck pipelines (part 19)	supported at 2.5–3σ (a-posteriori caveat stands)
Mirror selection rule (exact seed symmetry)	real sky, 240-axis scan (part 19)	bound, not detection — exact mirror seed excluded at these scales
E-mode echo at registered axes	real sky, conditional nulls (part 20)	open — underpowered today (32% vs 98% at LiteBIRD-class)
QC ceiling: n* ≈ 400 logical qubits per coherent block	part 15; industrial fault-tolerance roadmaps	open — no tension yet, decades of runway
No objective collapse (unitary engine)	macromolecule interferometry; Diósi–Penrose heating bounds	consistent so far

The pattern worth stating: the architecture is a class, not a point, and observation prunes it exactly as the constraints pruned rule space in §4 — observation has deleted the lattice substrate, the medium coupling, and the scalar metric, leaving sprinkle-like discreteness under a tensor metric with a quantum-or-bounded engine. Cornering, applied to ourselves. `[Class A for the entries; Class C for the pattern]`

12 What is not claimed

> THE UNEARNED RESIDUE
>
> **Quantum mechanics is not derived from beneath.** Bell, PBR, Kochen–Specker, and §7's interference deficit close that road; the reconstruction route (§4) corners QM from axioms but does not explain the axioms — the render reading of them is interpretation, not theorem.

> **The Standard Model is not derived.** Anomaly cornering fixes charges given the fermion content; nothing here selects SU(3)×SU(2)×U(1), three generations, or the Yukawa sector.
>
> **The seed's identity is unknowable in principle** (underdetermination); only its class — compressible, low-entropy — is evidenced, and §11's ceiling shows the class statement itself has limited empirical reach.
>
> **Installed components are flagged where they occur:** singlet statistics (§3), Born-rule sampling (§7), the Ryu–Takayanagi dictionary, the island rule, and the evaporating hole's line density (§6). Convergence of the constraint program may terminate at a universality class rather than a unique rule — consistent with the hypothesis, since the engine below the class is unobservable, but a ceiling nonetheless.

13 Objections and replies

Beyond the hidden-variables objection answered in §3:

"Unfalsifiable." The architecture is a framework, and frameworks are not falsifiable — their instantiations are, exactly as "field theory" is not falsifiable but particular field theories are. The instantiations here name their kill conditions: the seed-symmetry program fails if increasingly sensitive symmetry-class searches of primordial data stay null while covering the simple-generator class (§11's measured ceiling defines "covering"); the classical-engine variant fails if fault-tolerant quantum computation scales; medium-coupled gravity fails equivalence-principle tests — and §6 records that it *already has*. A program that states in advance what would kill it is entitled to the word hypothesis.

"The CMB significance is post hoc." Conceded, and structural: the anomalies were found by looking, and the Planck isotropy analyses decline to reject isotropy after look-elsewhere correction. The joint p of §11 quantifies coherence under a stated hypothesis, not discovery significance. The escape is not rhetoric but registration: the prediction now stands, in advance, for independent probes of the same modes — E-mode polarization foremost — where common origin predicts recurrence and flukes predict disappearance. The first in-repo sweep of the real maps (§11, "First light") makes both halves concrete: the joint statistic lands where the injections forecast, and the selection-rule scan reports its first bound rather than a claimed detection. The registered E-mode test has likewise had its first pass (§11, "The echo") — no echo, with the honest rider that today's polarization noise gave it only 32% power; registration means the axes wait, unchanged, for data that can decide.

"Idle ontology; Occam cuts it away." All interpretations of quantum mechanics predict identically; they earn their keep by steering research — decoherence theory and ER=EPR both began as interpretive seriousness. This frame steers: the survival law and the selection-rule search exist because it asked "what would a written seed leave behind?" It also purchases the Past Hypothesis — which standard cosmology leaves brute — at the price of one assumption about the algorithmic complexity of initial data. If the observational program nulls and quantum computing scales, the distinctive content shrinks toward interpretation; we state that in advance too.

"Rendered — for whom?" No viewer is required. The render layer is defined by the chart, a structural map between state spaces, not by minds; "rendering" names the output side of the map. Observers are patterns inside the render — §8 measures how cheaply they arise (21% of collisions leave a one-bit record) — they are not preconditions of it.

"This is reductionism renamed." Reductionism keeps observed adjacency fundamental: atoms are where their tables are. The distinctive claim here is *adjacency freedom* — the chart need not be an isometry — and that single liberty is what does the work in §3, §5, and every holographic result cited. Remove it and the program collapses into ordinary microphysics; that it does not collapse is the content.

14 Relation to prior work

The engine/render architecture is a synthesis, not an invention: digital physics (Zuse 1969; Fredkin; Wheeler 1989), the cellular-automaton interpretation ('t Hooft 2016), causal sets (Bombelli–Lee–Meyer–Sorkin 1987; Bombelli–Henson–Sorkin 2006), holography ('t Hooft 1993; Susskind 1995; Maldacena 1997), entanglement-derived geometry (Ryu–Takayanagi 2006; Van Raamsdonk 2010; Maldacena–Susskind ER=EPR 2013; Cao–Carroll–Michalakis 2017), entropic gravity (Jacobson 1995; Jacobson 2015), black-hole thermodynamics and analog gravity (Hawking 1974; Unruh 1981, 1995;

Corley–Jacobson 1996; Steinhauer 2016), the Page curve and entropy islands (Page 1993; Penington 2019; Almheiri–Engelhardt–Marolf–Maxfield 2019), the entanglement first law and its Einstein equivalence (Casini 2008; Blanco–Casini–Hung–Myers 2013; Faulkner–Guica–Hartman–Myers–Van Raamsdonk 2013), operational reconstructions (Hardy 2001; CDP 2011; Masanes–Müller 2011), envariance (Zurek 2003), decoherence (Zeh; Zurek), algorithmic thermodynamics (Zurek 1989), and the CMB anomaly literature (de Oliveira-Costa et al. 2004; Planck 2013–2019 isotropy papers). What this program adds is (i) a single runnable architecture in which these results cohere, (ii) the quantitative cornering measurements of §4, and (iii) the survival law and selection-rule observational program of §11.

Appendix A — reproducibility. All experiments are single-file Python (numpy + Pillow only) in github.com/joeleaver/the-rendered-universe, each running in seconds to ~1 minute on a laptop: `run.py` (laws & doors), `bell.py` (CHSH), `corner.py / census.py / frontier.py / ledger3d.py` (cornering), `entangle.py / curve.py` (geometry), `tempo.py / selfgrav.py / universal.py / ripple.py / shear.py / firstlaw.py / quench.py / horizon.py / page.py` (dynamics & gravity, through tensor gravitational waves, the entanglement first law and its time-dependent form, Hawking radiation, and the Page curve), `duality.py / eraser.py / nogo.py` (quantum layer), `arrow.py / insider.py` (arrow & observers), `sprinkle.py / chiral.py` (Lorentz & doubling), `axioms.py / generation.py` (QM & SM cornering), `collider.py` (axiom matrix), `fingerprint.py / sky.py / firstlight.py / echo.py` (the prediction program, ending on the real Planck temperature and polarization maps; the reduced real-sky data ship in-repo with URL + SHA-256 provenance, and `tools/fetch_sky.py / fetch_pol.py` rebuild them from the archives). Films and figures are generated by the scripts.

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